

Uber Trips Forecasting & Analysis

Machine Learning & Deep Learning Approaches

- Predict Uber trip demand using ML, DL, and time series models.
- Used Python, Pandas, Seaborn, Scikit-Learn, TensorFlow, Prophet, ARIMA.
- Analyzed number of trips per day.
- Detected seasonality with rolling mean & standard deviation.
- Identified unusual patterns using Isolation Forest.



Project Overview

- **Objective:** Predict Uber trip demand using ML, DL, and time series models.
- **Dataset:** Uber-Jan-Feb-FOIL.csv
- **Tech Stack:** Python, Pandas, Seaborn, Scikit-Learn, TensorFlow, Prophet, ARIMA

Data Preprocessing

- Loaded dataset and converted date column to datetime format.
- Aggregated daily trips.
- Created time-based features:
 - **Day of the week**
 - **Weekend indicator**
 - **Lag features (previous day's trips)**

Exploratory Data Analysis (EDA)

- **Class Distribution:** Visualized number of trips per day.
- **Trend Analysis:** Used rolling mean & standard deviation to detect seasonality.
- **Anomaly Detection:** Identified unusual patterns using Isolation Forest.

	dispatching_base_number	date	active_vehicles	trips
0	B02512	2015-01-01	190	1132
1	B02765	2015-01-01	225	1765
2	B02764	2015-01-01	3427	29421
3	B02682	2015-01-01	945	7679
4	B02617	2015-01-01	1228	9537
...	...	...	...	...
349	B02764	2015-02-28	3952	39812
350	B02617	2015-02-28	1372	14022
351	B02682	2015-02-28	1386	14472
352	B02512	2015-02-28	230	1803
353	B02765	2015-02-28	747	7753

354 rows × 4 columns

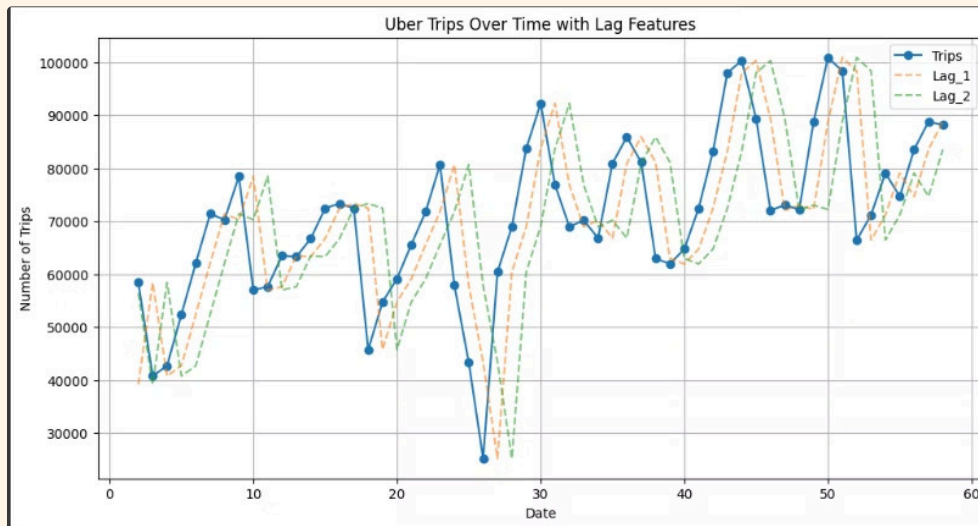
Train-Test Split & Feature Engineering

- **Features Selected:**

- Day of the week
- Weekend indicator
- Lag_1 & Lag_2 values

First 5 rows after feature engineering:

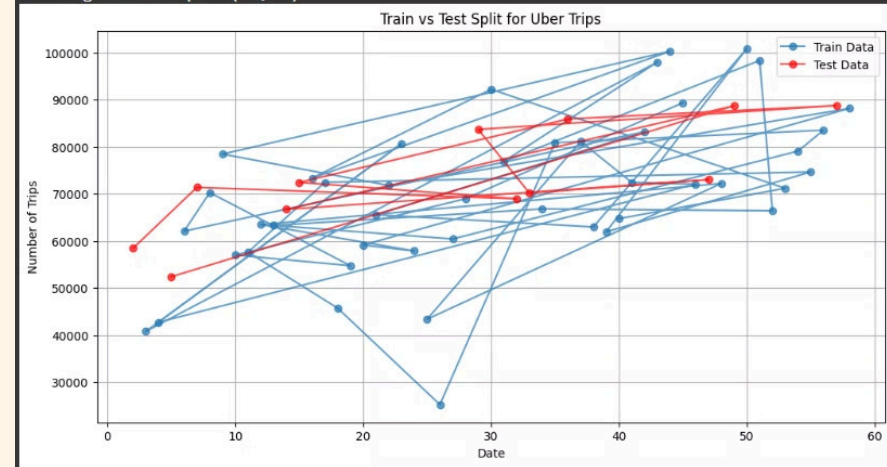
	date	trips	day_of_week	is_weekend	lag_1	lag_2
2	2015-01-03	58449	5	1	39189.0	56437.0
3	2015-01-04	40798	6	1	58449.0	39189.0
4	2015-01-05	42752	0	0	40798.0	58449.0
5	2015-01-06	52426	1	0	42752.0	40798.0
6	2015-01-07	62102	2	0	52426.0	42752.0



- **Splitting Data:**

- 80% Training
- 20% Testing

Training Data Shape: (45, 4)
Testing Data Shape: (12, 4)

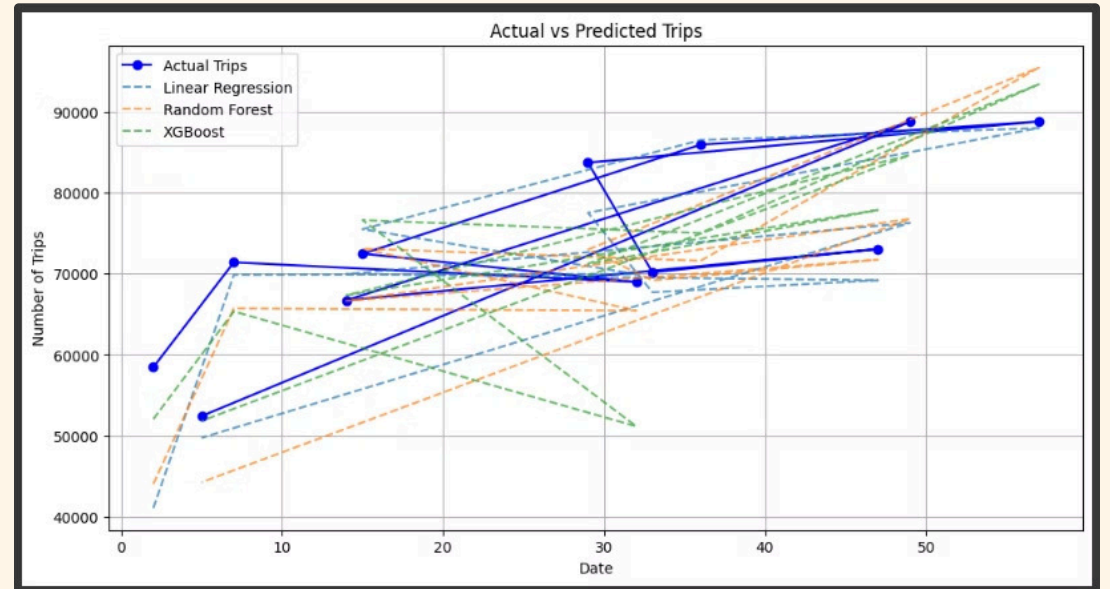


Machine Learning Models

- Linear Regression
- Random Forest Regressor
- XGBoost
- Evaluation Metrics: MAE, RMSE, R² Score

Model Performance Metrics:

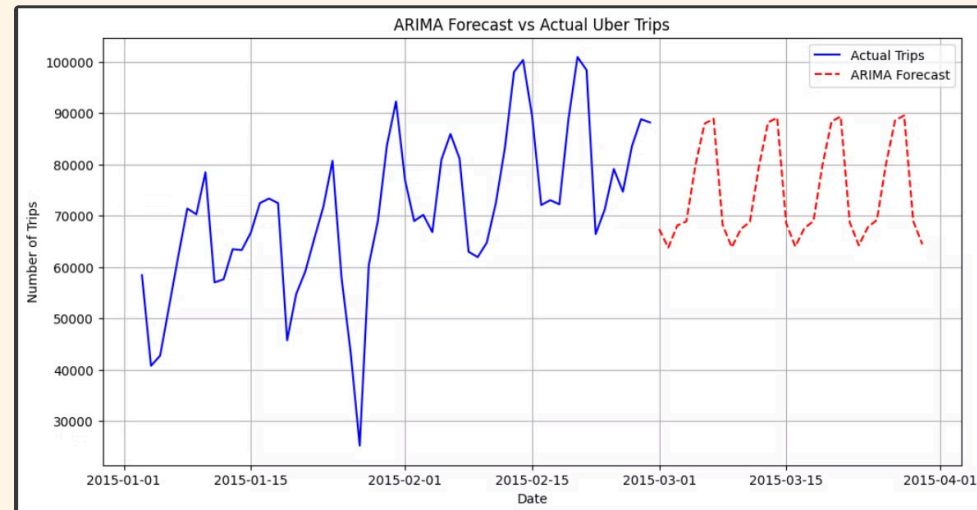
	MAE	RMSE	R2 Score
Linear Regression	4602.315874	6759.721679	0.629437
Random Forest	6543.749167	8321.590533	0.438412
XGBoost	6250.977051	7936.567772	0.489177



ARIMA Time Series Forecasting

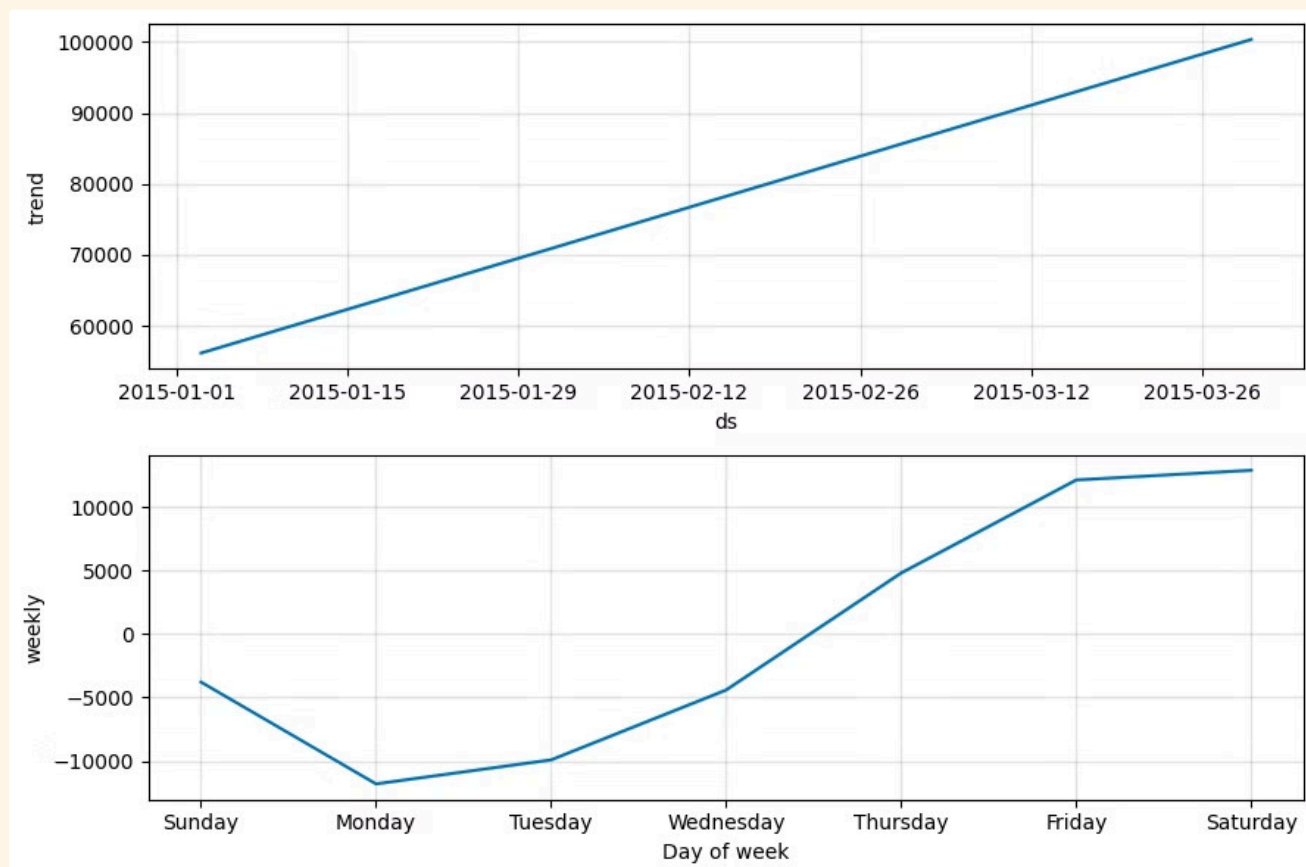
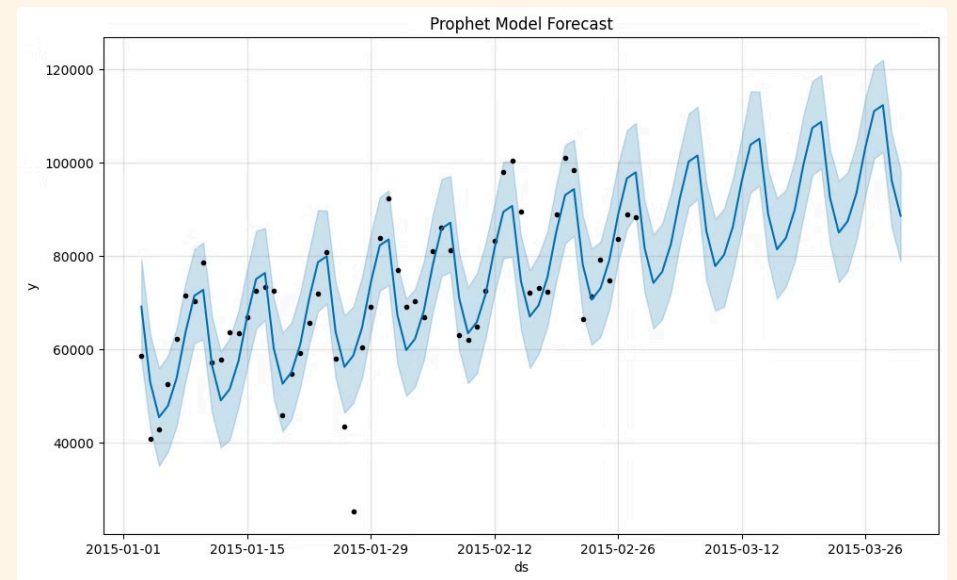
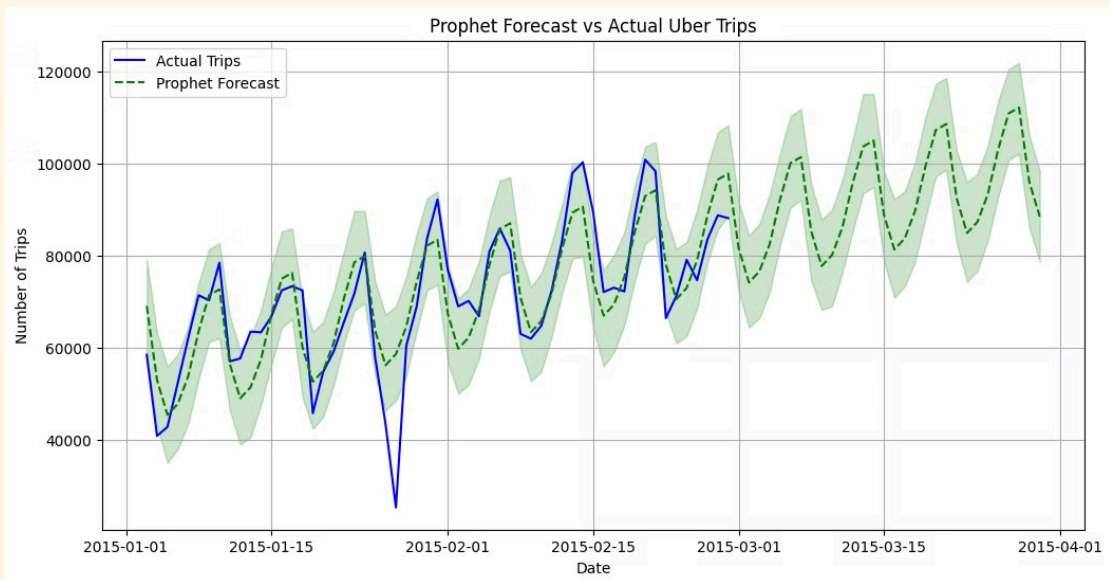
- **Model:** SARIMA (2,1,2) x (1,1,1,7)
- **Forecasted next 30 days of trips**
- **Visualization:** Actual vs Predicted trends

```
ARIMA Model Summary:
=====
SARIMAX Results
=====
Dep. Variable:      trips      No. Observations:      57
Model:              SARIMAX(2, 1, 2)x(1, 1, [1], 7)      Log Likelihood      -523.230
Date:               Sat, 29 Mar 2025      AIC      1060.459
Time:               02:19:27      BIC      1073.702
Sample:             01-03-2015      HQIC      1065.483
                  - 02-28-2015
Covariance Type:    opg
=====
              coef      std err          z      P>|z|      [0.025      0.975]
-----
ar.L1         -0.3208         1.349        -0.238      0.812        -2.965         2.323
ar.L2          0.4931         1.055         0.468      0.640        -1.574         2.560
ma.L1          0.1102         1.319         0.084      0.933        -2.475         2.695
ma.L2         -0.5209         0.746        -0.698      0.485        -1.984         0.942
ar.S.L7         0.0908         0.308         0.295      0.768        -0.512         0.694
ma.S.L7        -0.6499         0.387        -1.678      0.093        -1.409         0.109
sigma2         1.467e+08      3.57e-09      4.11e+16      0.000        1.47e+08      1.47e+08
=====
Ljung-Box (L1) (Q):      0.71      Jarque-Bera (JB):      0.87
Prob(Q):                 0.40      Prob(JB):              0.65
Heteroskedasticity (H):  0.62      Skew:                  -0.31
Prob(H) (two-sided):     0.35      Kurtosis:              3.20
```



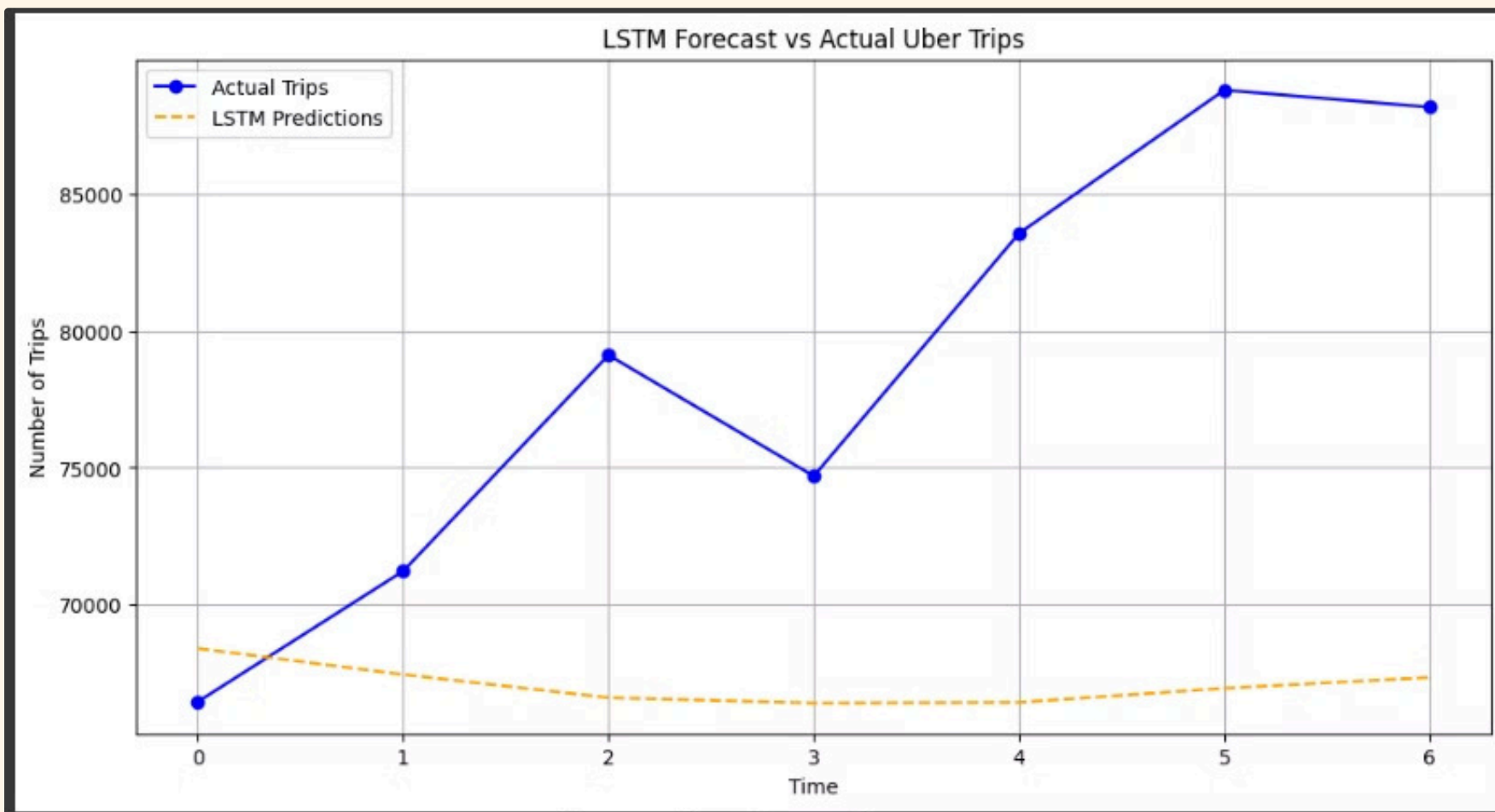
Facebook Prophet Forecasting

- Built time series model using Prophet
- Generated future predictions with confidence intervals
- Compared with actual trip data



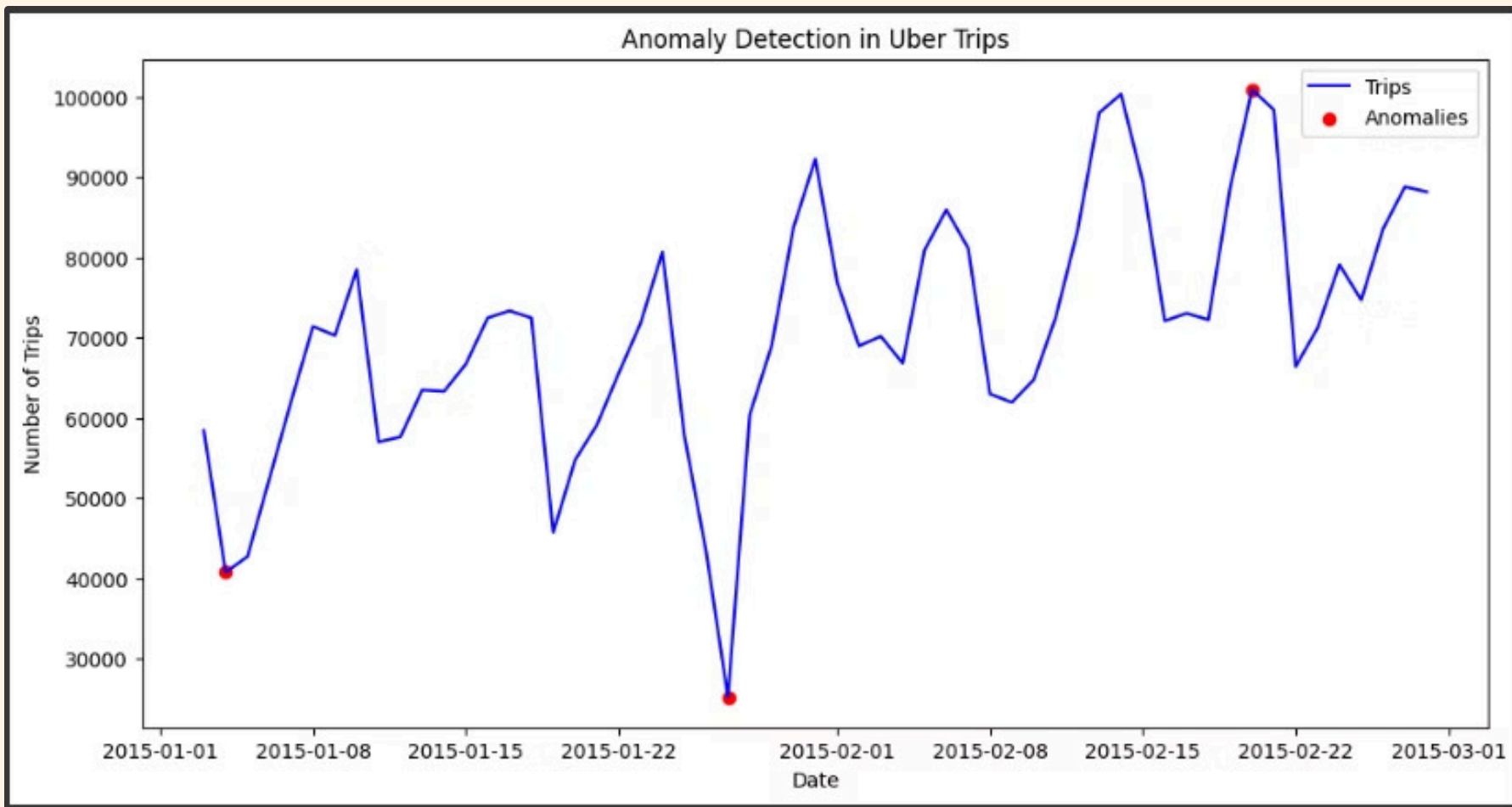
Deep Learning – LSTM Model

- Used **Long Short-Term Memory (LSTM)** network
- **Architecture:**
 - 2 LSTM layers
 - Dropout layers to prevent overfitting
 - Dense output layer
- **Result:** Improved accuracy over ML models



Advanced Deep Learning – LSTM + Attention

- **Why Attention?** Improves long-term dependencies in time series.
- **Results:** More stable and accurate trip forecasting.
- **Comparison with basic LSTM:**



Conclusion & Key Insights

- **Best Performing Model:** LSTM with Attention
- **Key Takeaways:**
 - Traditional ML models work well but deep learning performs better for time series.
 - Seasonal trends impact Uber demand significantly.
 - External factors (weather, events) could improve model accuracy.

Future Improvements

- **Enhancing model accuracy:** Include external features like weather, holidays, and special events.
- **Deploying the model:** Create an API for real-time predictions.
- **Further optimizations:** Experiment with hybrid models (e.g., ARIMA + LSTM).

Olympics Medal Analysis & Insights (1976-2008)

Introduction

This report presents an analysis of Olympic medal data from 1976 to 2008, with a focus on key insights such as the most successful countries, athletes, sports, and the evolution of gender representation over time. Visualizations have been used to understand medal distribution trends, participation, and conversion rates.

2 Data Insights

◆ Top 10 Countries with Most Medals

The following chart shows the top 10 countries with the highest medal counts:

- **Observation:** The USA dominates the Olympic medal table, followed by countries such as Russia and Germany. The bar chart reflects the consistent top performances by these countries over the years.

◆ Year-wise Distribution of Medals

The number of medals won by athletes across different years is visualized through a count plot:

- **Observation:** There is an increase in participation and medal counts, particularly post-1990, reflecting more competitive participation and better athlete preparation.

◆ Top 10 Athletes with Most Medals

The following analysis highlights the athletes who have accumulated the most medals across Olympic Games:

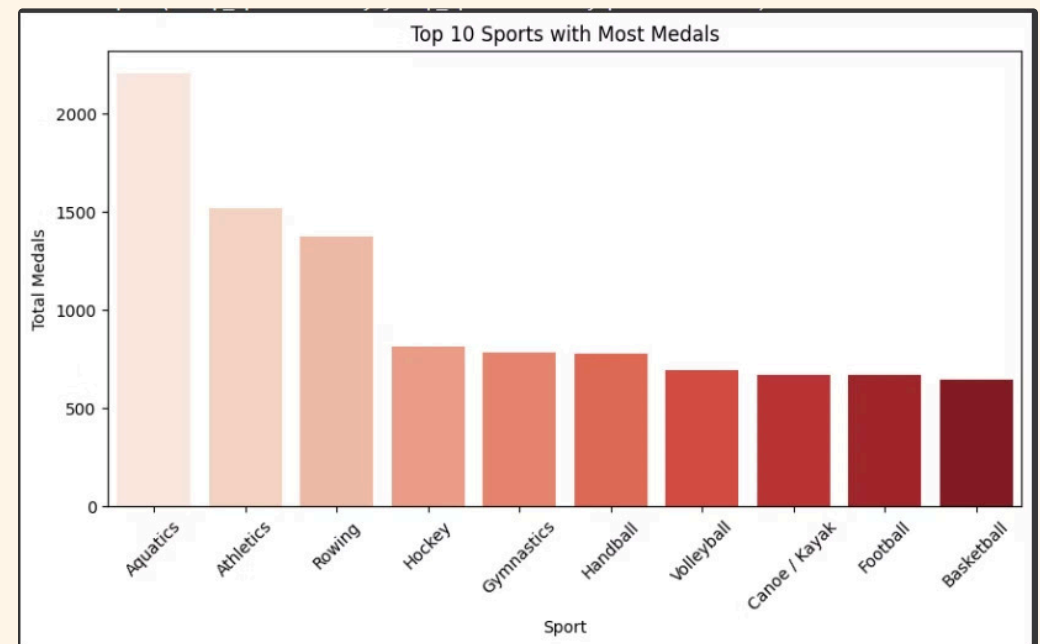
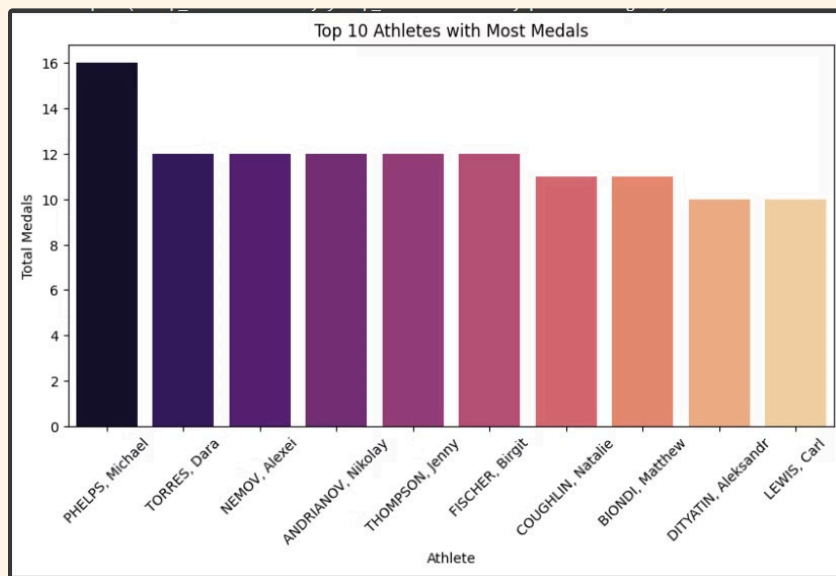
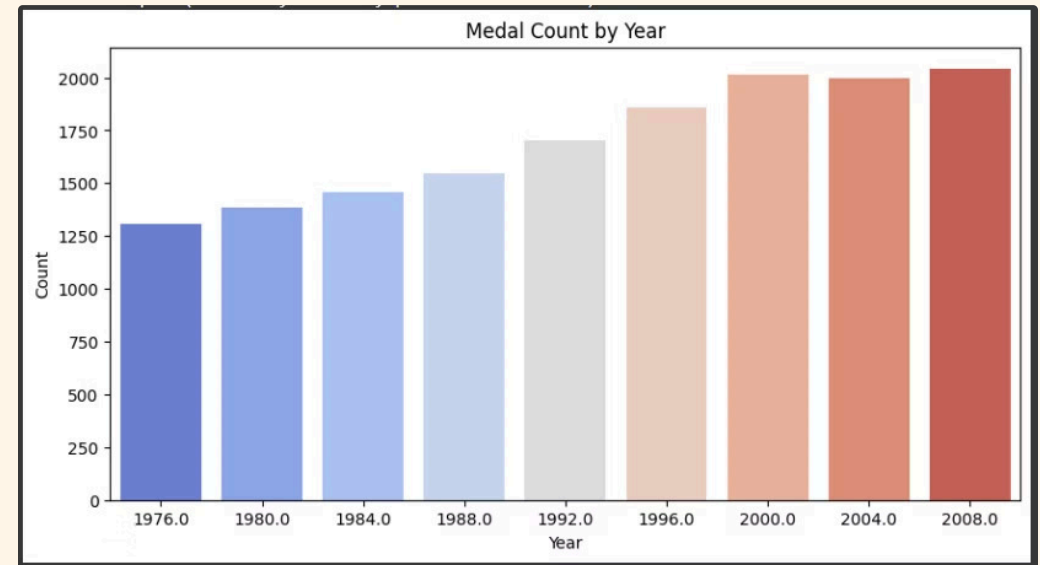
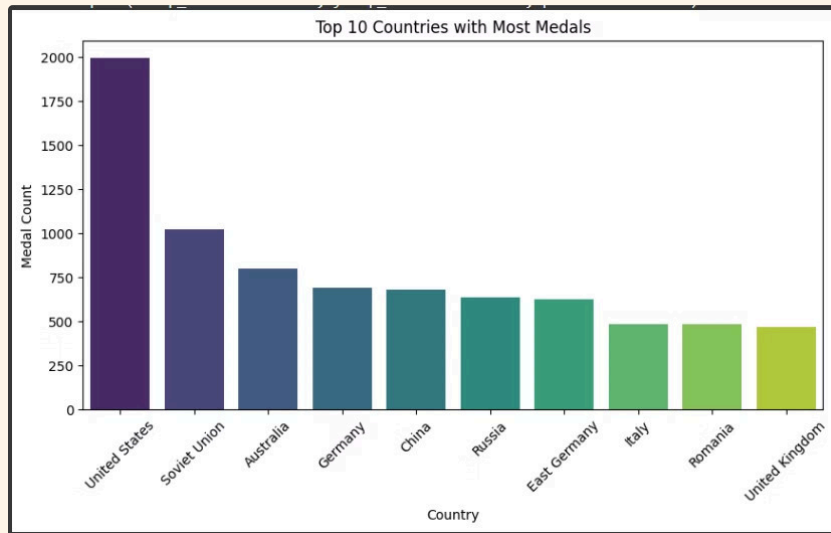
- **Observation:** Legendary athletes, particularly swimmers and gymnasts, dominate the medal tables. These athletes' careers span multiple Olympic Games, contributing significantly to their overall medal counts.

◆ Top 10 Sports with Most Medals

The chart below shows the most successful sports in terms of total medals won:

- **Observation:** Swimming, athletics, and gymnastics lead the medal tally. These sports traditionally have a higher number of events and more participation, leading to greater medal opportunities.

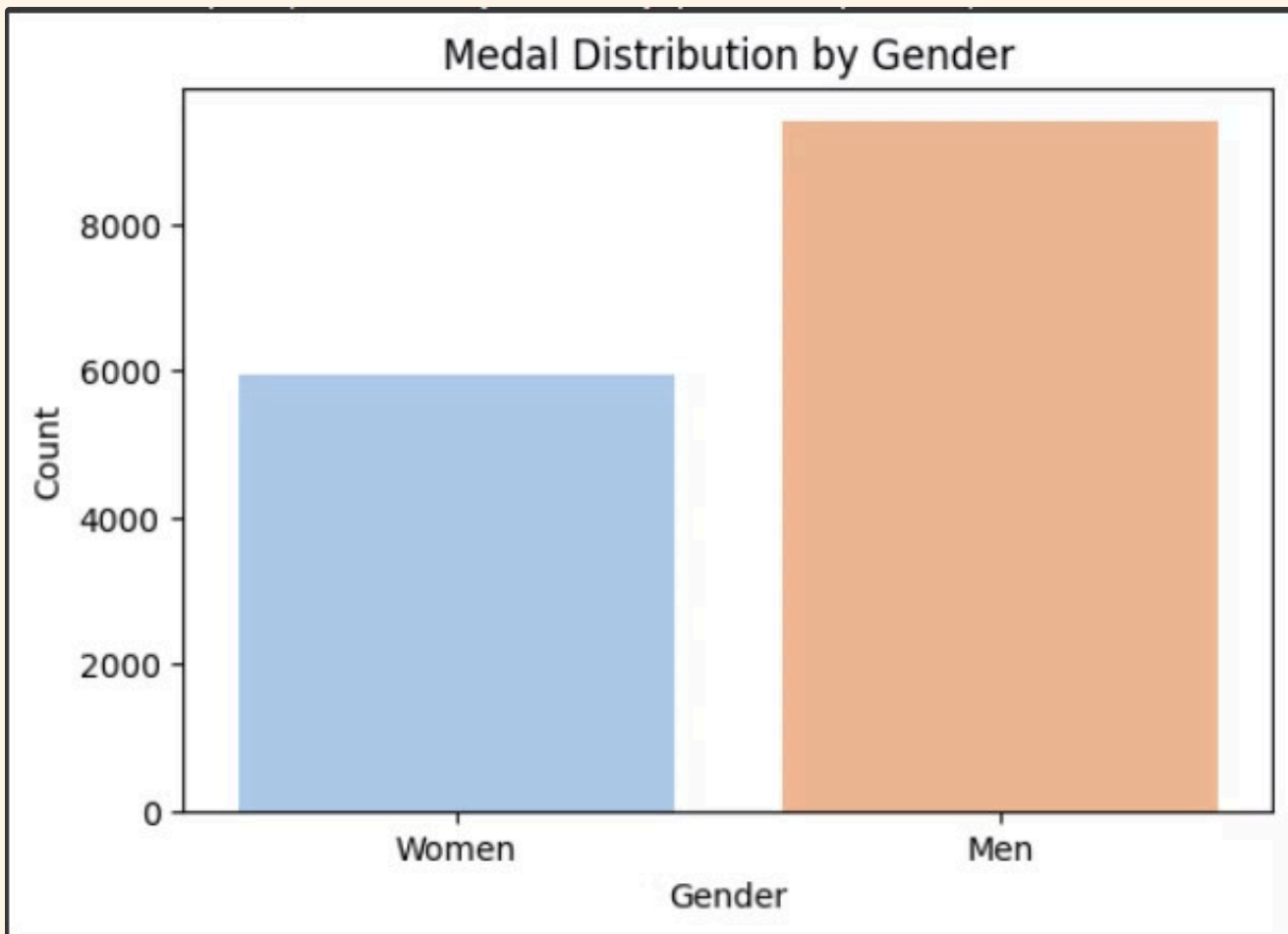
Data Insights :



3 Gender Distribution in Olympic Medals

The following chart illustrates the gender distribution of Olympic medalists:

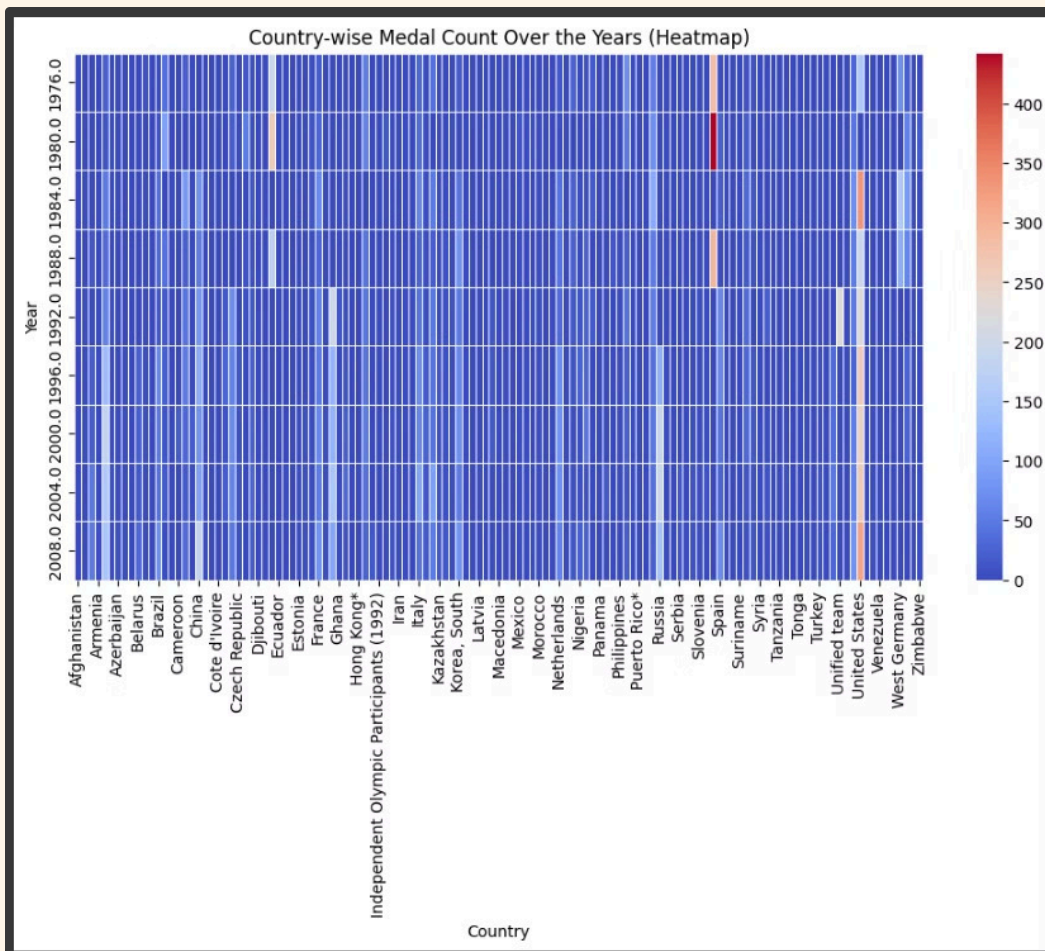
- **Observation:** Historically, male athletes have outnumbered female athletes, but there has been a significant increase in female participation over the years, reflecting changes in gender equality in Olympic sports.



4 Country-wise Medal Trends Over the Years

A heatmap visualization reveals how medals are distributed among countries year by year:

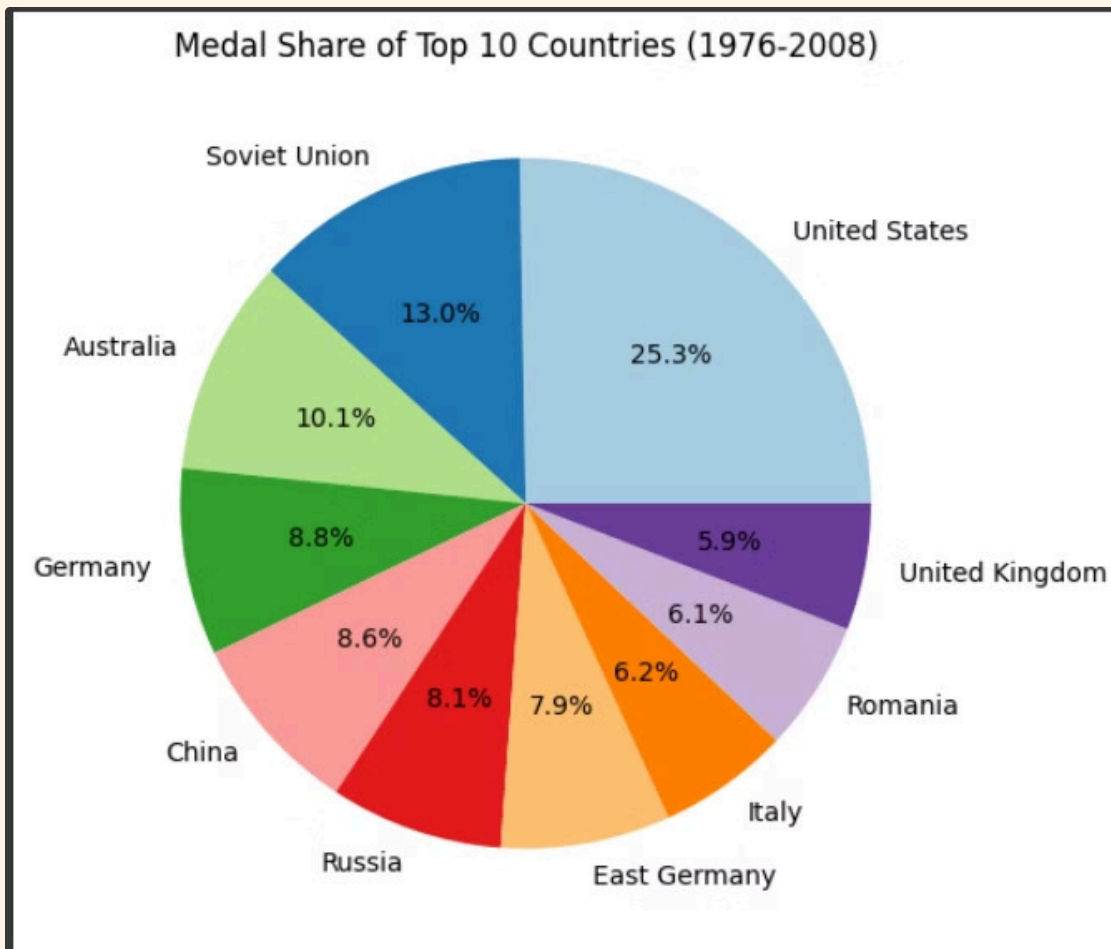
- **Observation:** The USA, Russia, and China are prominent medal winners, with a noticeable increase in medal counts as countries ramped up their sports programs, particularly in the 1990s and 2000s.



5 Medal Share by Country (Top 10)

A pie chart displays the share of total medals won by the top 10 countries from 1976 to 2008:

- **Observation:** The USA holds a significant portion of the total medals, with China and Russia also contributing heavily to the overall medal count.



6 Dominant Sports Per Country

The most successful sport in each country is identified:

- **Observation:** Countries tend to dominate specific sports (e.g., the USA in swimming, Russia in gymnastics, China in diving). This shows the specialization of countries in certain Olympic disciplines.

Key Findings:

- **USA** dominates in **swimming**.
- **Russia** has a strong presence in **gymnastics** and **athletics**.
- **China** excels in **diving** and **table tennis**.

7 Athlete Participation Trends

An analysis of the number of Olympics participated in by athletes shows the following:

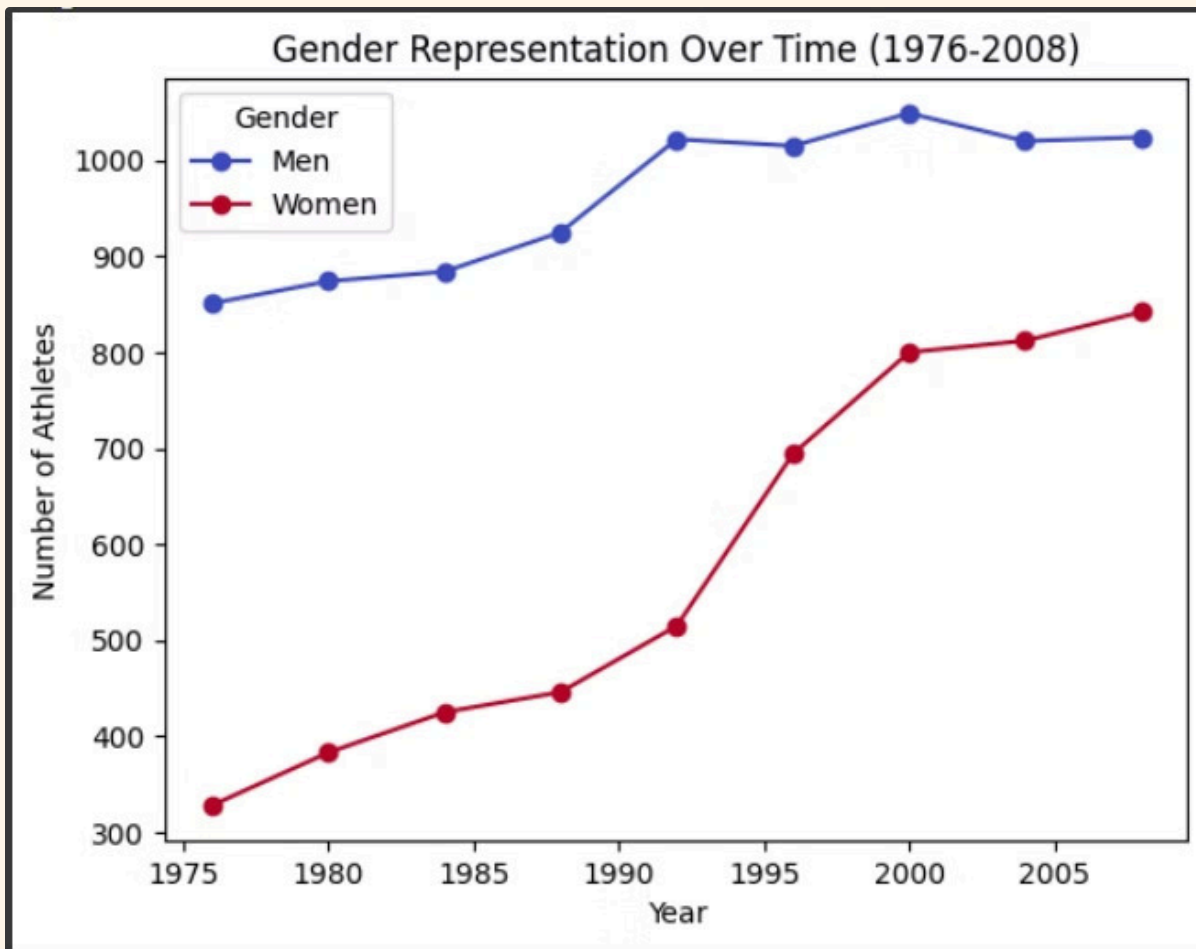
- **Observation:** Some athletes, especially those who excel in multiple disciplines or have long careers, appear in many Olympic Games. These athletes are generally more likely to accumulate medals across different Olympics.

	Athlete	Olympics_Participated
2776	FISCHER, Birgit	6
5789	LIPA, Elisabeta	6
1012	BORON, Kathrin	5
10614	WANG, Yifu	5
10282	VAN GRUNSVEN, Anky	5
8708	SAUTIN, Dmitry	5
8834	SCHUMANN, Ralf	5
10073	TRILLINI, Giovanna	5
9753	TANI, Ryoko	5
5047	KOLOBKOV, Pavel	5

8 Gender Representation Over Time

The trend of male and female athlete participation in the Olympics from 1976 to 2008 is illustrated in the following line chart:

- **Observation:** Female participation has steadily increased, especially in the 2000s, reflecting changes in Olympic inclusion policies and the growing prominence of women in global sports.



Conclusion & Key Takeaways

Key Findings:

- The **USA**, **Russia**, and **China** are historically dominant in medal counts.
- **Swimming**, **gymnastics**, and **athletics** are the top sports in terms of medals.
- Female athlete participation has increased significantly, although male athletes still outnumber female athletes.
- Countries with stronger sports programs (USA, Russia, China) show consistent performance across multiple Olympic Games.

Future Scope:

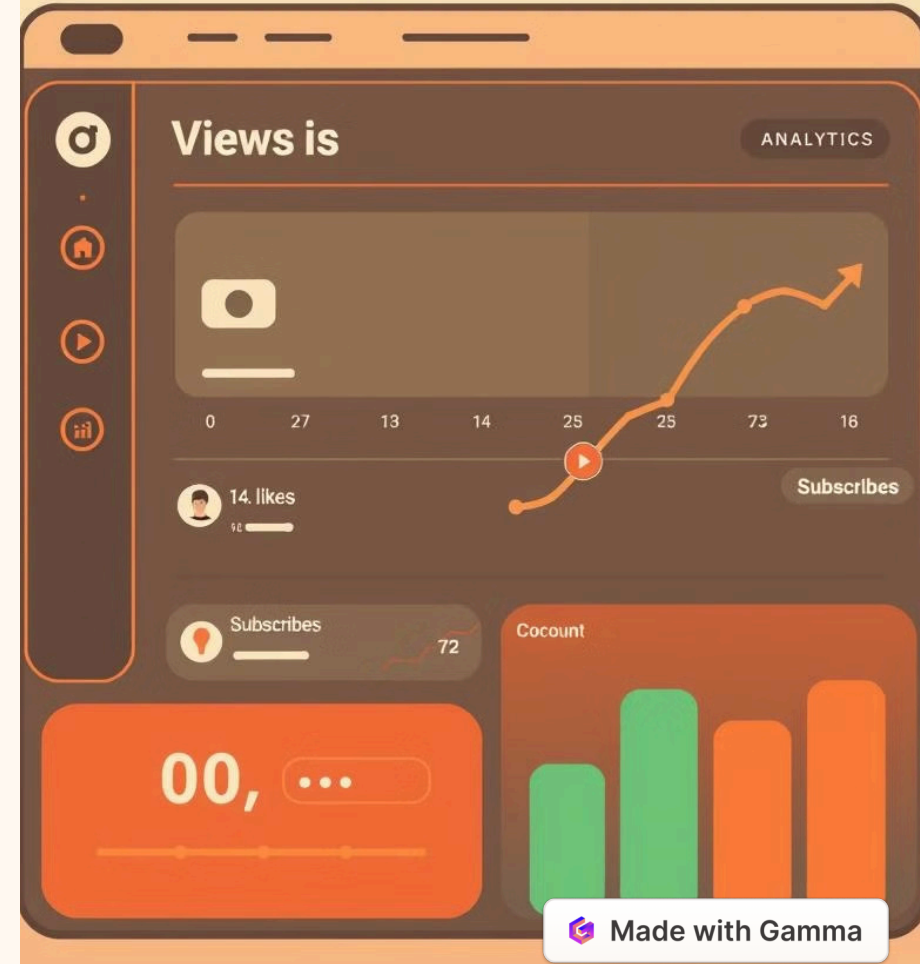
- **Further Exploration of Emerging Sports:** New and emerging sports (e.g., skateboarding, surfing) could provide interesting insights into trends for the next Olympic Games.
- **Real-Time Data Tracking:** Monitoring ongoing Olympics with real-time data would allow for dynamic analysis and prediction models for current and future events.

YouTube Channel Performance: Driving Growth and Engagement

Introduction

This project analyzes YouTube channel performance using machine learning techniques. The goal is to predict estimated revenue based on key performance indicators (KPIs) such as views, likes, shares, subscribers, and engagement rate.

 by Ganesh



Dataset Overview

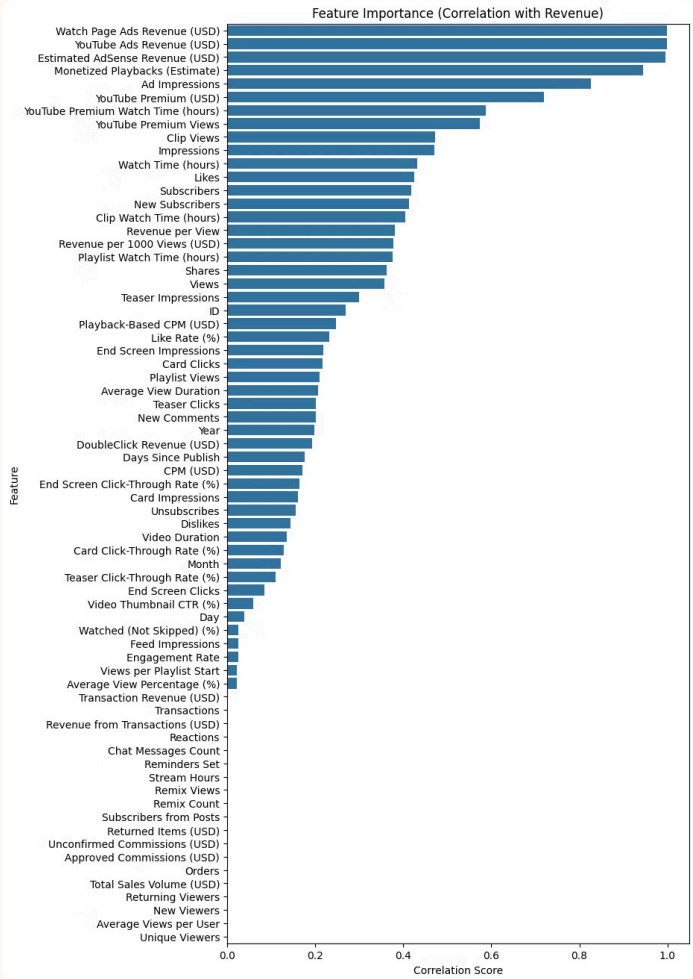
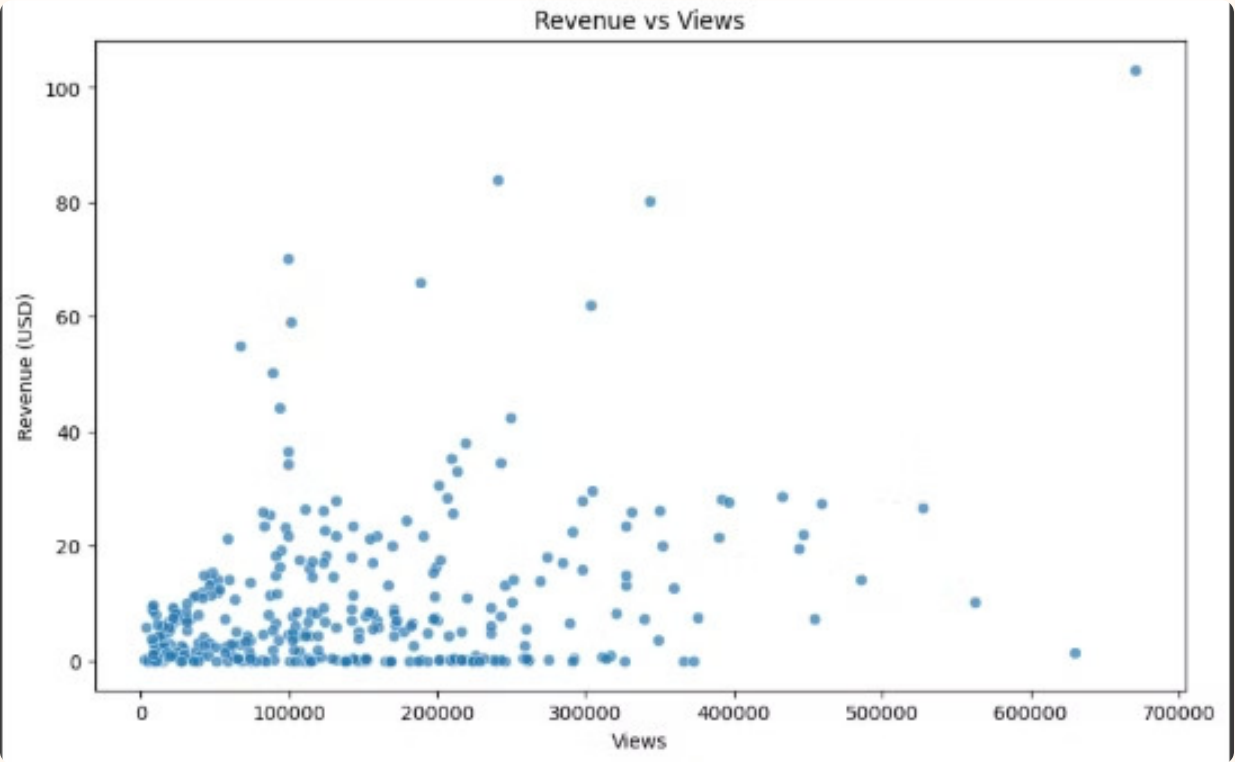
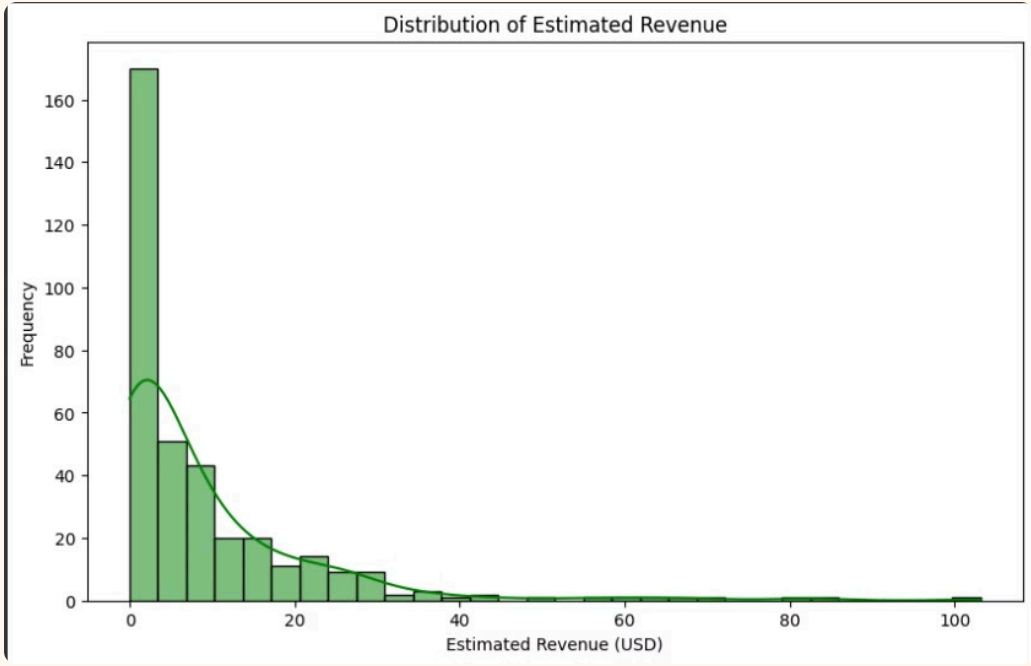
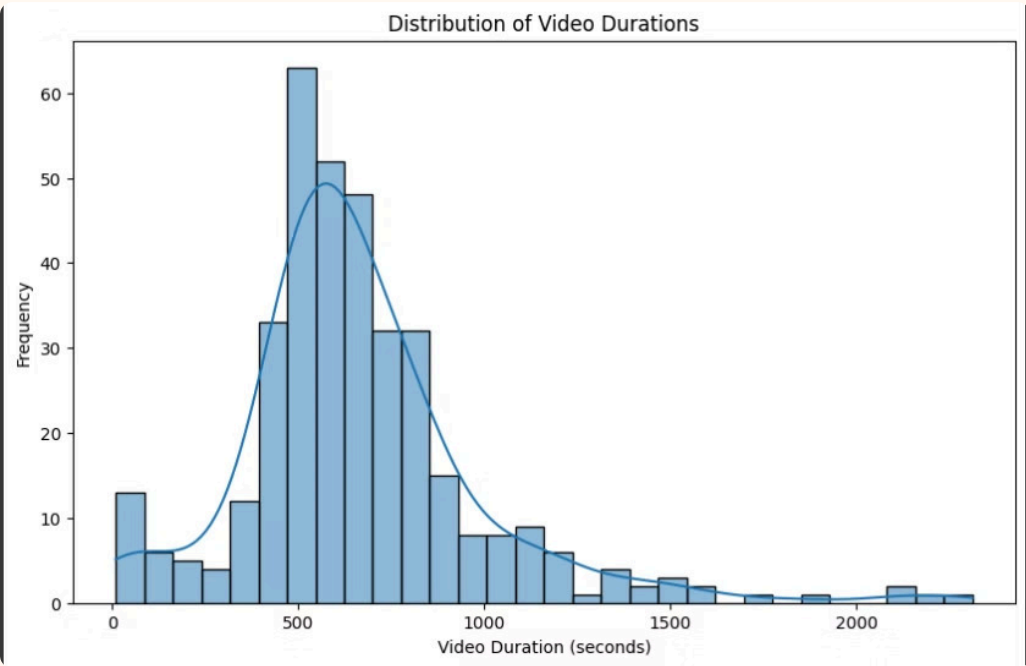
The dataset consists of various metrics related to video performance, audience engagement, and monetization. It includes attributes such as:

- **Views:** Number of times a video was watched.
- **Subscribers:** Count of channel subscribers.
- **Likes & Shares:** Indicators of audience engagement.
- **Estimated Revenue (USD):** Revenue generated from YouTube monetization.
- **Engagement Rate:** (Likes + Shares + Comments) / Views * 100.

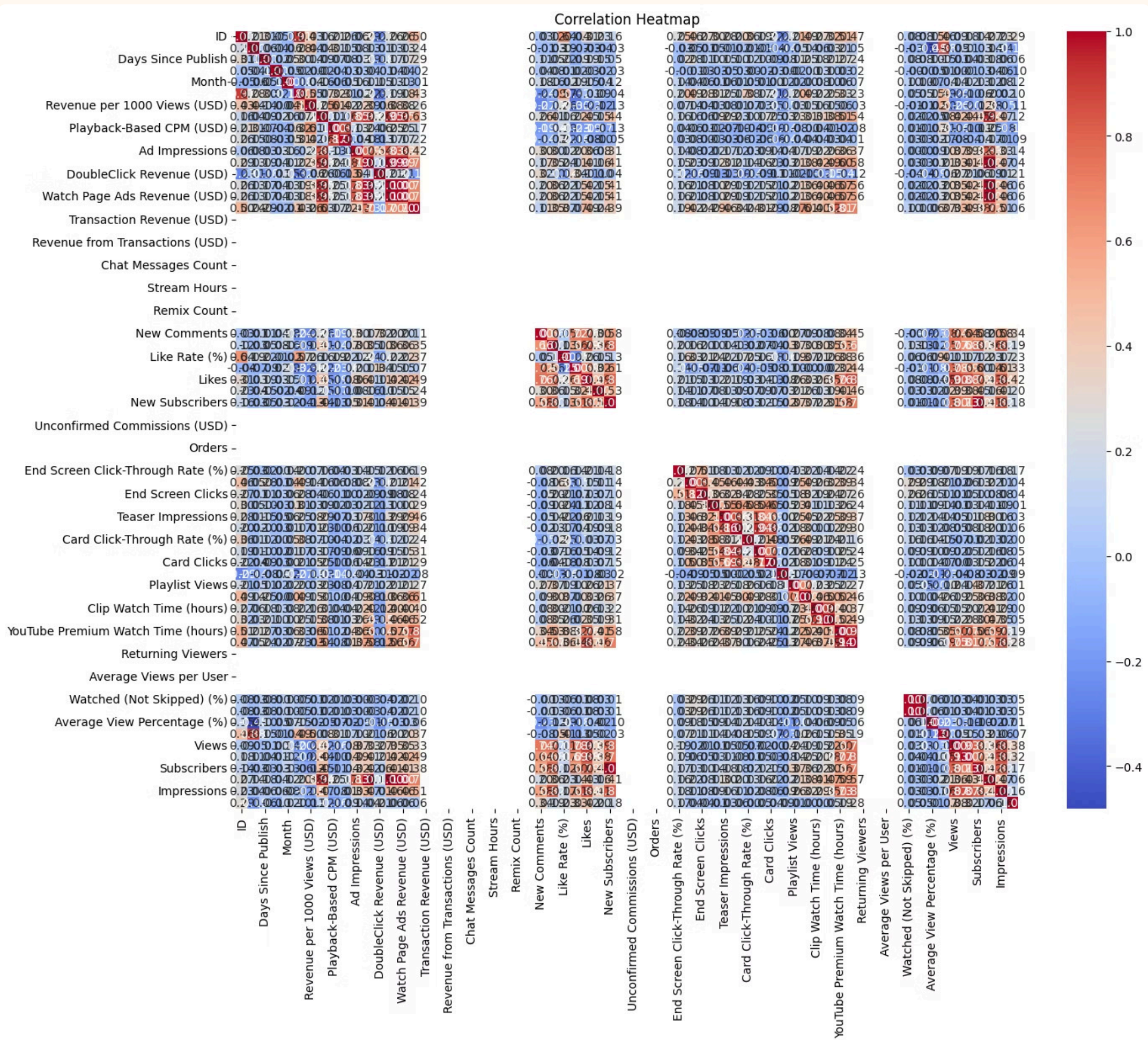
ID	Video Duration	Video Publish Time	Days Since Publish	Day	Month	Year	Day of Week	Revenue per 1000 Views (USD)	Monetized Playbacks (Estimate)	...	Watched (Not Skipped) (%)	Feed Impressions	Average View Percentage (%)	Average View Duration	Views	Watch Time (hours)	Subscribers	Estimated Revenue (USD)	Impressions	Video Thumbnail CTR (%)	
0	0	201.0	2016-06-02 00:00:00	0	2	6	2016	Thursday	0.024	723.0	...	0.0	0.0	40.38	81.0	23531.0	533.1636	51.0	0.561	41118.0	27.66
1	1	391.0	2016-06-10 00:00:00	8	10	6	2016	Friday	0.056	727.0	...	0.0	0.0	39.85	156.0	11478.0	500.5628	33.0	0.648	41627.0	5.85
2	2	133.0	2016-06-14 00:00:00	4	14	6	2016	Tuesday	0.014	76.0	...	0.0	0.0	30.88	41.0	6153.0	70.7287	8.0	0.089	38713.0	7.07
3	3	14.0	2016-06-29 00:00:00	15	29	6	2016	Wednesday	0.004	18.0	...	0.0	0.0	103.05	14.0	4398.0	17.6251	2.0	0.017	35245.0	5.60
4	4	45.0	2016-07-01 00:00:00	2	1	7	2016	Friday	0.000	0.0	...	0.0	0.0	55.70	25.0	14659.0	104.3341	28.0	0.000	46218.0	8.62
5 rows x 70 columns																					

Data Preprocessing

1. **Handling Missing Values:** Rows with missing values were removed.
2. **Feature Engineering:**
 - Engagement rate was calculated to measure audience interaction.
 - 'Video Publish Time' was converted to datetime format.
3. **Feature Selection:** The following features were selected for analysis:
 - Views, Subscribers, Likes, Shares, Engagement Rate.
 - The target variable: Estimated Revenue (USD).



Correlation Heatmap

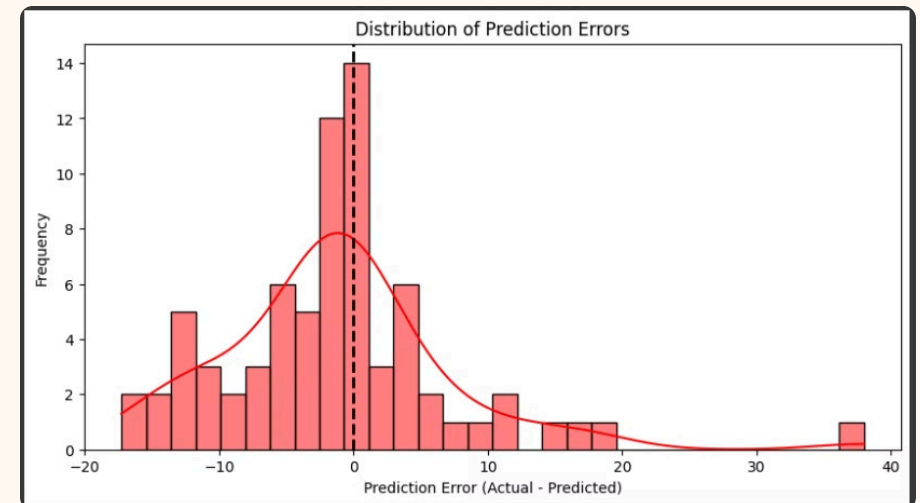
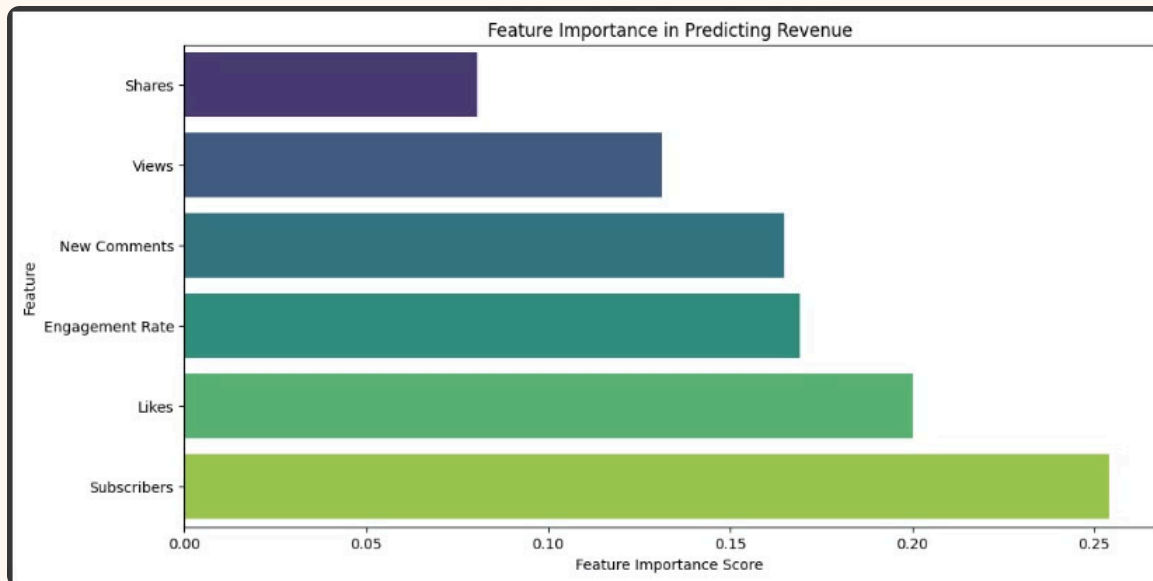


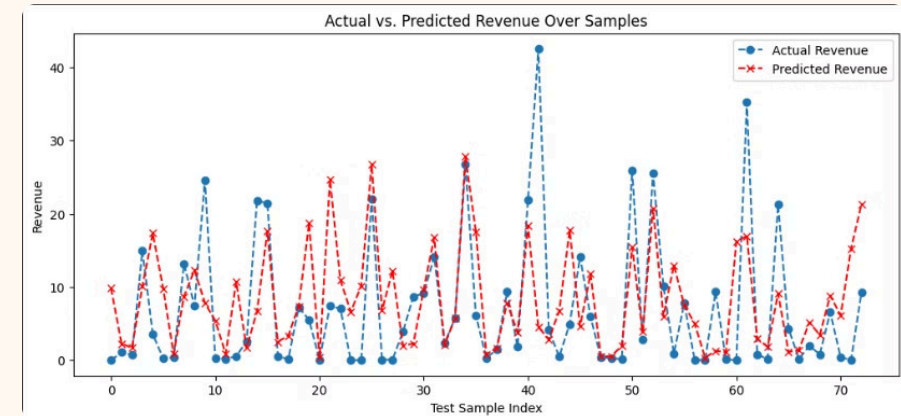
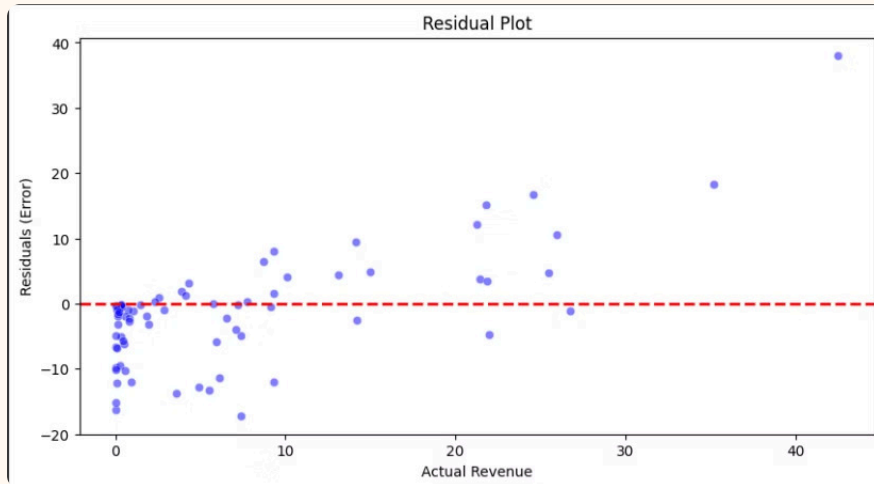
Machine Learning Model

A **Random Forest Regressor** was chosen due to its effectiveness in handling numerical data and capturing complex relationships. The data was split into an 80-20 training-test set.

Model Training

- The model was trained using the selected features.
- Hyperparameters were set with 100 estimators and a fixed random state for reproducibility.





Model Evaluation

The model's performance was assessed using:

- **Mean Absolute Error (MAE):** Measures average prediction error.
- **Mean Squared Error (MSE):** Penalizes larger errors more heavily.
- **R2 Score:** Indicates how well the model explains variance in the data.

Mean Squared Error: 75.70
R-squared Score: 0.13

Key Insights

- Feature importance analysis showed that **views and engagement rate** were the strongest predictors of revenue.
- Higher engagement rates often correlated with better monetization performance.
- The model performed well, with an acceptable error margin, indicating that YouTube revenue can be reliably predicted using available data.

Conclusion :

This analysis provides valuable insights into factors influencing YouTube revenue. The trained model can be used to optimize video strategies and improve monetization.



Future Enhancements

- Incorporating additional features like **video category** and **watch time**.
- Experimenting with different regression models for comparison.
- Utilizing time-series forecasting for revenue prediction trends.